ApexPlanet Cybersecurity Internship - Task 1 FullReport

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- **Task 1** : **Foundation & Environment Setup**

Introduction :

This report covers Task 1: Foundation and Environment Setup. The objective is to understand
cybersecurity fundamentals, create a virtual lab environment, learn Linux basics, networking
concepts, cryptography fundamentals, and security tools.

Step 1 : **Cybersecurity Basics**

❖ CIA Triad (Confidentiality, Integrity, Availability)

The **CIA Triad** is a fundamental concept in cybersecurity that helps organizations protect information and information systems. It consists of three key security principles:

1. Confidentiality

Confidentiality means ensuring that sensitive information is accessible only to authorized individuals and is protected from unauthorized access.

Objectives

- Prevent unauthorized disclosure of information.

- Protect personal, financial, and business data.
- Maintain privacy and secrecy of sensitive information.

Methods to Achieve Confidentiality

- **Encryption:** Converts data into unreadable form for unauthorized users.
- **Passwords and Authentication:** Verify user identity before granting access.
- **Access Control:** Allows only authorized users to access specific data.

Example

A bank customer's account details should only be accessible to the customer and authorized bank employees. If a hacker gains access to these details, confidentiality is compromised.

Threats to Confidentiality

- Hacking
- Data breaches
- Phishing attacks
- Insider threats
- Unauthorized access

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2. Integrity

Integrity refers to maintaining the accuracy, consistency, and trustworthiness of data throughout its lifecycle.

Objectives

- Ensure data remains accurate and complete.
- Prevent unauthorized modification of information.
- Detect any changes made to data.

Methods to Achieve Integrity

- **Hashing:** Verifies that data has not been altered.
- **Digital Signatures:** Confirm authenticity and integrity of data.
- **Version Control:** Tracks changes to files and documents. .

Example

A student's examination marks stored in a university database should not be changed by unauthorized persons. Any alteration must be properly authorized and recorded.

Threats to Integrity

- Malware and viruses
- Unauthorized modifications
- Human errors
- Database corruption
- Cyberattacks

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3. Availability

Availability ensures that information and systems are accessible to authorized users whenever needed.

Objectives

- Keep systems operational and accessible.
- Minimize downtime.
- Ensure business continuity.

Methods to Achieve Availability

- **Regular Backups:** Recover data in case of failure.
- **Redundant Systems:** Use backup servers and network connections.
- **Disaster Recovery Plans:** Restore operations after incidents.

Example

An online shopping website should remain accessible to customers 24/7. If the server crashes and customers cannot access the website, availability is affected.

Threats to Availability

- Hardware failures
- Power outages
- Denial-of-Service (DoS) attacks
- Natural disasters
- Software bugs

- ❖ Explore Threat Types: Phishing, Malware, DDoS, SQL Injection, Brute Force, Ransomware.

Cyber threats are malicious activities designed to damage, disrupt, steal, or gain unauthorized access to computer systems, networks, and data. Some of the most common cyber threats are discussed below.

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1. Phishing

Phishing is a cyberattack in which attackers impersonate trusted organizations or individuals to trick users into revealing sensitive information such as passwords, credit card details, or personal data.

Characteristics:

- Fake emails, messages, or websites.
- Creates a sense of urgency.
- Attempts to steal login credentials or financial information.

Example:

A user receives an email claiming to be from a bank asking them to click a link and verify their account details.

Prevention:

- Verify sender identities.
- Avoid clicking suspicious links.
- Use multi-factor authentication (MFA).

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2. Malware

Malware (Malicious Software) refers to any software designed to damage, disrupt, or gain unauthorized access to computer systems.

Types of Malware:

- Virus
- Worm
- Trojan Horse
- Spyware
- Adware

Example:

A user downloads an infected file that installs spyware and secretly records keystrokes.

Prevention:

- Install antivirus software.
- Keep software updated.
- Avoid downloading files from untrusted sources.

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3. DDoS (Distributed Denial of Service)

A **DDoS attack** occurs when multiple compromised devices send massive amounts of traffic to a server or website, overwhelming it and making it unavailable to legitimate users.

Characteristics:

- Uses a large number of infected devices (botnet).
- Causes service interruptions.
- Affects website performance and availability.

Example:

An online shopping website crashes during a sale because attackers flood it with fake traffic.

Prevention:

- Use DDoS protection services.
- Implement traffic filtering.
- Monitor network activity.

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4. SQL Injection

SQL Injection (SQLi) is a web application attack where attackers insert malicious SQL commands into input fields to manipulate a database.

Objectives:

- Access confidential data.
- Modify or delete database records.
- Bypass authentication mechanisms.

Example:

An attacker enters malicious code into a login form to gain access without a valid password.

Prevention:

- Use parameterized queries.
- Validate user inputs.
- Apply secure coding practices.

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5. Brute Force Attack

A **Brute Force Attack** is a trial-and-error method used to guess passwords or encryption keys by trying many possible combinations.

Characteristics:

- Automated attack.
- Targets weak passwords.
- Can take seconds or years depending on password strength.

Example:

An attacker uses software to test thousands of password combinations on an email account.

Prevention:

- Use strong passwords.
- Enable account lockout policies.
- Implement multi-factor authentication.

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6. Ransomware

Ransomware is a type of malware that encrypts a victim's files or locks their system and demands payment for restoration.

Characteristics:

- Encrypts important data.
- Demands ransom payment.
- Can spread across networks.

Example:

A company finds all its files encrypted and receives a message demanding payment in cryptocurrency.

Prevention:

- Regularly back up data.
- Keep systems updated.
- Avoid opening suspicious attachments.

❖ Study Attack Vectors: Social Engineering, Wireless Attacks, Insider Threats.

An **attack vector** is the method or pathway used by attackers to gain unauthorized access to a system, network, or data. Understanding attack vectors helps organizations strengthen their cybersecurity defenses.

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1. Social Engineering

Social Engineering is a technique in which attackers manipulate people into revealing confidential information or performing actions that compromise security.

Characteristics:

- Exploits human psychology rather than technical vulnerabilities.
- Relies on trust, fear, curiosity, or urgency.
- Often used to steal passwords, financial information, or sensitive data.

Common Types:

- **Phishing:** Fake emails or messages.
- **Pretexting:** Creating a false identity or scenario.
- **Baiting:** Offering something attractive to lure victims.
- **Tailgating:** Gaining physical access by following authorized personnel.

Example:

An attacker calls an employee pretending to be IT support and asks for login credentials.

Prevention:

- Security awareness training.
- Verify identities before sharing information.
- Use multi-factor authentication (MFA).

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2. Wireless Attacks

Wireless Attacks target wireless networks such as Wi-Fi to intercept data, gain unauthorized access, or disrupt communications.

Characteristics:

- Exploit weak wireless security settings.
- Can be performed remotely within signal range.
- May lead to data theft or network compromise.

Common Types:

- **Evil Twin Attack:** Fake Wi-Fi hotspot mimicking a legitimate network.
- **Packet Sniffing:** Capturing wireless data packets.
- **Wi-Fi Password Cracking:** Breaking weak passwords.
- **Man-in-the-Middle (MITM) Attack:** Intercepting communication between users and networks.

Example:

A hacker creates a fake public Wi-Fi network in a café to steal users' login credentials.

Prevention:

- Use strong Wi-Fi encryption (WPA3/WPA2).
- Avoid connecting to unknown public networks.
- Use VPNs on public Wi-Fi.

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3. Insider Threats

Insider Threats come from individuals within an organization who misuse their authorized access to harm systems, data, or operations.

Characteristics:

- Involves employees, contractors, or business partners.
- Can be intentional or accidental.

- Difficult to detect because insiders already have access.

Types:

- **Malicious Insider:** Intentionally steals or damages data.
- **Negligent Insider:** Causes security incidents through carelessness.
- **Compromised Insider:** Account is taken over by an external attacker.

Example:

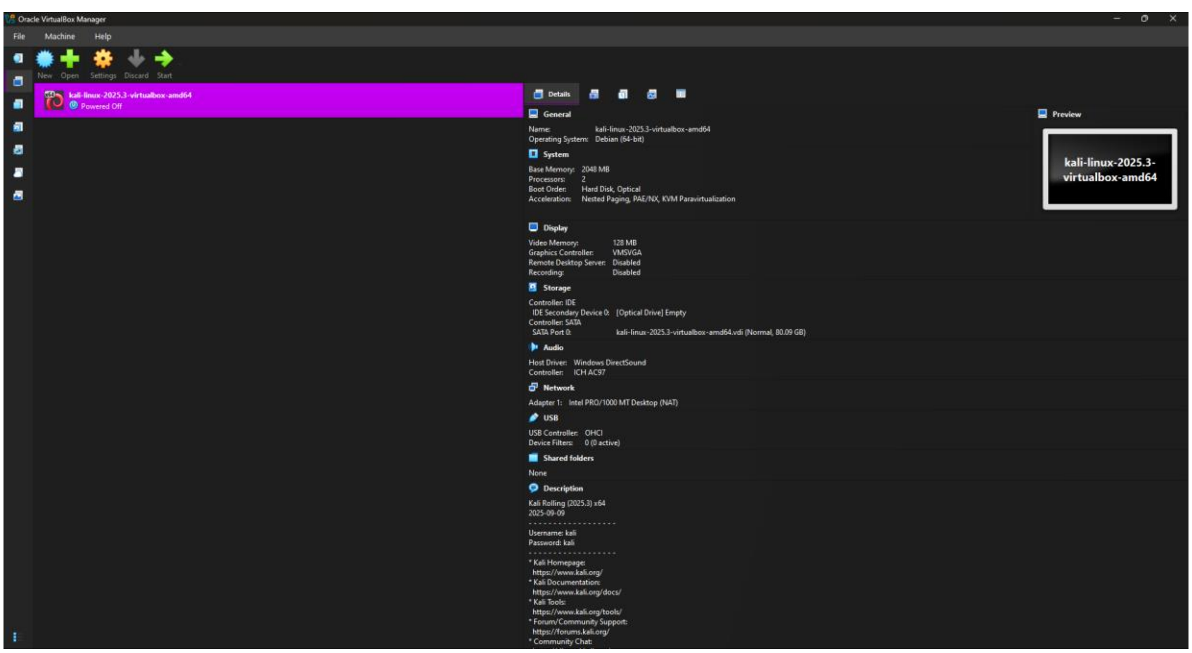
An employee copies confidential company data and shares it with competitors.

Prevention:

- Apply the principle of least privilege.
- Monitor user activities.
- Conduct regular security training.
- Implement access controls and audits.

Step 2 : Lab Environment Setup

❖ Install VirtualBox or VMware



- Kali Linux was installed on VirtualBox and configured as the attacker machine for cybersecurity testing and network analysis.

❖ Install Kali Linux as the attacker machine.

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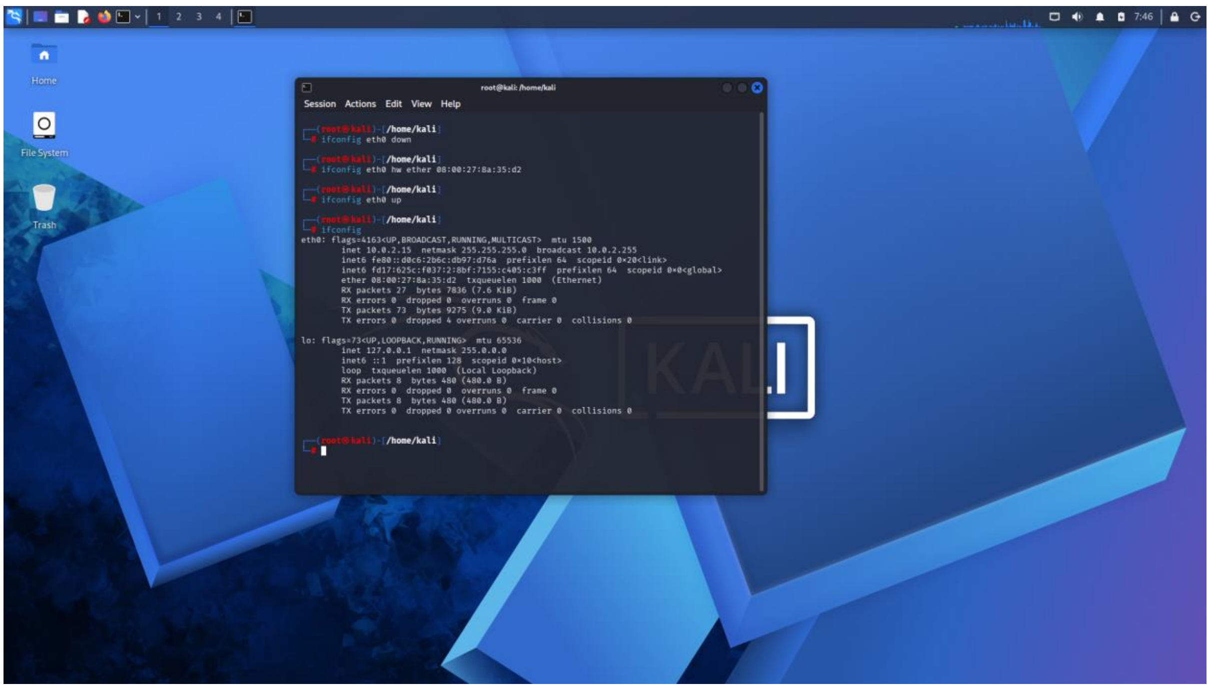
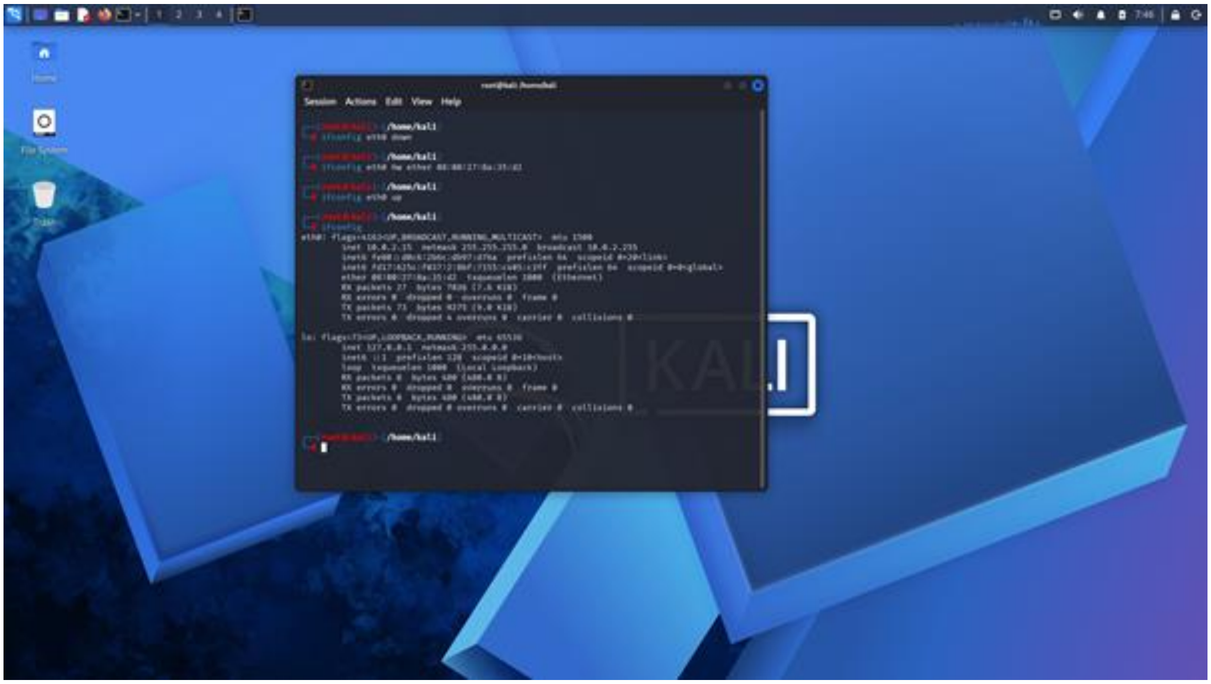
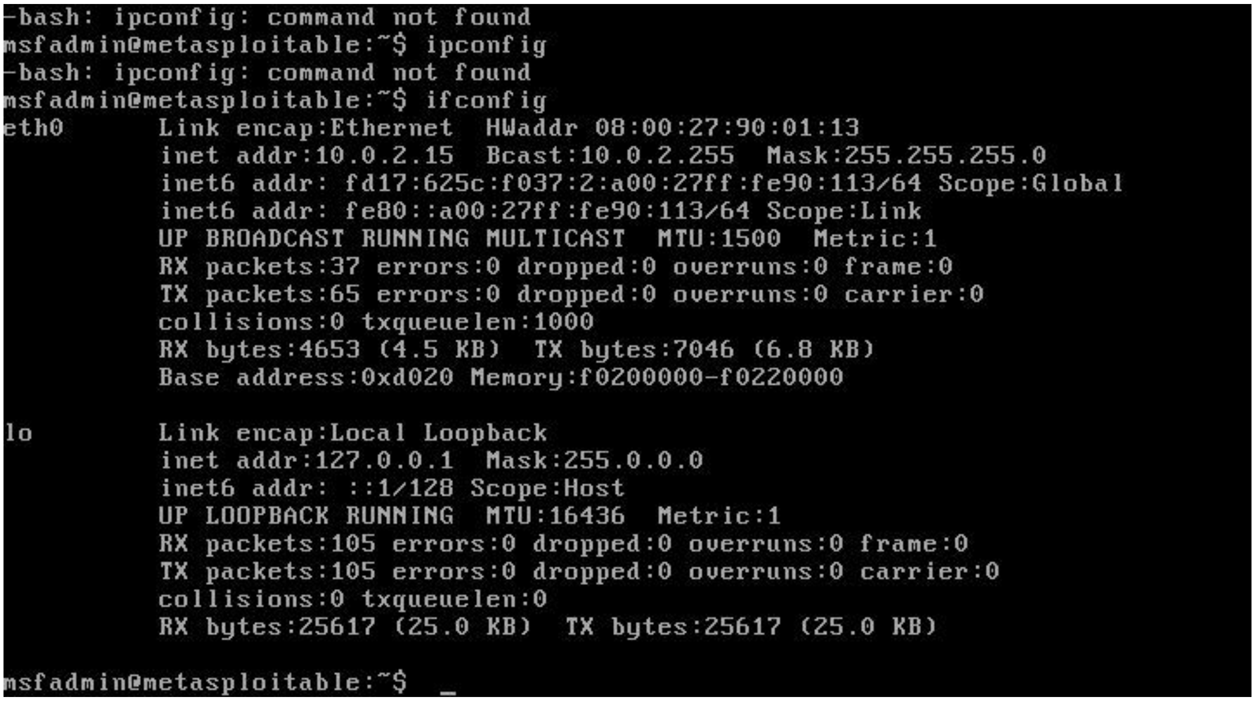


❖ Install Metasploitable2 as target machines.



➢ Configure a private lab network (Host-Only Adapter).



```
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ ipconfig
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:90:01:13
          inet addr:10.0.2.15  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fd17:625c:f037:2:a00:27ff:fe90:113/64 Scope:Global
          inet6 addr: fe80::a00:27ff:fe90:113/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:37 errors:0 dropped:0 overruns:0 frame:0
          TX packets:65 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4653 (4.5 KB)  TX bytes:7046 (6.8 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:105 errors:0 dropped:0 overruns:0 frame:0
          TX packets:105 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:25617 (25.0 KB)  TX bytes:25617 (25.0 KB)

msfadmin@metasploitable:~$ _
```

VirtualBox was successfully installed and configured on the system. Kali Linux and Metasploitable 2 virtual machines were set up to create a cybersecurity lab environment. The virtual machines were connected using the appropriate network settings for testing and analysis.

Step 3 : Linux Fundamentals

❖File System Navigation (cd, ls, pwd).

File System Navigation (`cd`, `ls`, `pwd`)

File System Navigation refers to the process of moving through directories (folders) and locating files within an operating system using terminal commands. In Linux, the most basic commands used for navigation are `pwd`, `ls`, and `cd`.

---

1. `pwd` (Print Working Directory)

What does it do?

The `pwd` command displays the full path of the directory you are currently working in.

Syntax:

```
pwd
```

Example:

```
$ pwd
/home/kali
```

Explanation:

The output indicates that the current working directory is `/home/kali`.

---

2. `ls` (List)

What does it do?

The `ls` command lists the files and directories contained in the current directory.

Syntax:

```
ls
```

Example:

```
$ ls
Desktop   Documents   Downloads   Pictures
```

Explanation:

This output shows that the current directory contains four folders: **Desktop, Documents, Downloads, and Pictures**.

Useful Options:

Display detailed information:

```
ls -l
```

Display hidden files and directories:

```
ls -a
```

---

3. `cd` (Change Directory)

What does it do?

The `cd` command is used to move from one directory to another.

Syntax:

```
cd directory_name
```

Example:

```
$ cd Documents
```

Explanation:

This command changes the current directory to the **Documents** folder.

Common Uses:

Move to the parent directory:

```
cd ..
```

Return to the home directory:

```
cd
```

Move to the root directory:

```
cd /
```

Practical Example

Suppose you open the terminal and type:

```
$ pwd
/home/kali
```

Check the contents of the current directory:

```
$ ls
Desktop  Documents  Downloads
```

Move into the Documents directory:

```
$ cd Documents
```

Verify your new location:

```
$ pwd
/home/kali/Documents
```

Conclusion

- **`pwd`** – Displays the current working directory.
- **`ls`** – Lists the files and directories in the current directory.
- **`cd`** – Changes the current directory to another location.

These commands are fundamental Linux commands and are essential for navigating and managing the file system through the terminal.

❖File & Directory Permissions (chmod, chown).

File & Directory Permissions (`chmod`, `chown`)

File and Directory Permissions in Linux determine who can read, write, or execute a file or directory. They help protect files from unauthorized access and ensure that only the appropriate users can modify or use them. The two main commands used to manage permissions and ownership are **`chmod`** and **`chown`**.

---

1. `chmod` (Change Mode)

What does it do?

The `chmod` command is used to change the permissions of a file or directory.

Syntax:

```
chmod permissions file_name
```

Permission Types:

Linux permissions are represented by three symbols:

- **r** = Read permission
- **w** = Write permission
- **x** = Execute permission

These permissions are assigned to three categories:

- **User (u)** – The owner of the file
- **Group (g)** – Users belonging to the file's group
- **Others (o)** – All other users

Example:

```
chmod u+x script.sh
```

Explanation:

This command gives the file owner (`u`) execute (`x`) permission on the file `script.sh`.

Numeric Permission Method

Permissions can also be represented using numbers:

Permission	Value
Read (r)	4
Write (w)	2
Execute (x)	1

Example:

```
chmod 755 script.sh
```

Explanation:

- Owner: 7 (4+2+1 = read, write, execute)
- Group: 5 (4+1 = read, execute)
- Others: 5 (4+1 = read, execute)

This allows the owner full access while others can only read and execute the file.

---

2. `chown` (Change Ownership)

What does it do?

The `chown` command changes the owner and group ownership of a file or directory.

Syntax:

```
chown owner file_name
```

Example:

```
sudo chown kali report.txt
```

Explanation:

This command changes the ownership of `report.txt` to the user **kali**.

Changing Owner and Group Together

Syntax:

```
sudo chown owner:group file_name
```

Example:

```
sudo chown kali:developers project.txt
```

Explanation:

This command changes:

- The owner to **kali**
- The group to **developers**

for the file `project.txt`.

Practical Example

Suppose a file named `notes.txt` exists.

Check its permissions:

```
ls -l notes.txt
```

Output:

```
-rw-r--r-- 1 kali kali 120 Jun 14  notes.txt
```

Make the file executable:

```
chmod +x notes.txt
```

Change its owner:

```
sudo chown user1 notes.txt
```

Check the updated details:

```
ls -l notes.txt
```

Conclusion

- **chmod** is used to change the permissions of files and directories.
- **chown** is used to change the ownership of files and directories.
- Proper use of these commands helps maintain security, control access, and manage files effectively in Linux systems.

❖Package Management (apt, dpkg).

Package Management (`apt`, `dpkg`)

Package Management in Linux is the process of installing, updating, removing, and managing software packages. In Debian-based distributions such as **Ubuntu** and **Kali Linux**, the most commonly used package management tools are **`apt`** and **`dpkg`**.

---

1. `apt` (Advanced Package Tool)

What does it do?

The `apt` command is a high-level package management tool used to install, update, upgrade, and remove software packages from online repositories.

Syntax:

```
apt [command] package_name
```

Common `apt` Commands

Update the package list:

```
sudo apt update
```

Explanation:

Downloads the latest information about available packages from the configured repositories.

Upgrade installed packages:
```
sudo apt upgrade
```

Explanation:

Updates all installed packages to their newest available versions.

Install a package:
```
sudo apt install nmap
```

Explanation:

Installs the **Nmap** package along with its required dependencies.

Remove a package:
```
sudo apt remove nmap
```

Explanation:

Removes the package but may leave its configuration files behind.

Remove a package completely:
```
sudo apt purge nmap
```

Explanation:

Removes both the package and its configuration files.

Clean unused packages:
```
sudo apt autoremove
```

Explanation:

Deletes packages that were installed automatically but are no longer needed.

---

2. `dpkg` (Debian Package)

What does it do?

The `dpkg` command is a low-level package management tool used to install, remove, and inspect `.deb` package files directly.

Syntax:

```
dpkg [option] package_file.deb
```

Common `dpkg` Commands

Install a `.deb` package:

```
sudo dpkg -i package.deb
```

Explanation:

Installs the specified Debian package file.

Remove an installed package:

```
sudo dpkg -r package_name
```

Explanation:

Removes the package while keeping its configuration files.

List installed packages:

```
dpkg -l
```

Explanation:

Displays all packages currently installed on the system.

Show package information:

```
dpkg -s package_name
```

Explanation:

Provides detailed information about a specific installed package.

Practical Example

Suppose you want to install **Nmap**:

Update the package database:

```
sudo apt update
```

Install Nmap:

```
sudo apt install nmap
```

Check whether Nmap is installed:

```
dpkg -s nmap
```

View the package details:

```
dpkg -l | grep nmap
```

Difference Between `apt` and `dpkg`

Feature	`apt`	`dpkg`
Type	High-level package manager	Low-level package manager
Source	Uses online repositories	Uses local `.deb` files
Dependency Handling	Automatically resolves dependencies	Does not resolve dependencies automatically
Main Use	Installing, updating, and removing packages	Managing individual `.deb` packages

---

Conclusion

- **`apt`** is used for managing software from repositories and automatically handles dependencies.
- **`dpkg`** is used for working directly with `.deb` package files installed on the local system.
- Together, these tools provide an efficient way to install, update, and manage software packages in Debian-based Linux distributions such as Ubuntu and Kali Linux.

❖Networking Commands (ifconfig, ping, netstat, traceroute).

Networking Commands (`ifconfig`, `ping`, `netstat`, `traceroute`)

Networking commands are used in Linux to view network configurations, test connectivity, monitor network connections, and trace the path that data packets take across a network. Some of the most commonly used networking commands are **`ifconfig`**, **`ping`**, **`netstat`**, and **`traceroute`**.

---

1. `ifconfig` (Interface Configuration)

What does it do?

The `ifconfig` command is used to display and configure network interface settings, such as IP addresses and network adapters.

Syntax:

```
ifconfig
```

Example:

```
$ ifconfig
```

Sample Output:

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>
        inet 192.168.1.10
        netmask 255.255.255.0
```

Explanation:

This output shows information about the network interface (`eth0`), including its IP address and subnet mask.

---

2. `ping`

What does it do?

The `ping` command checks whether another device or website is reachable over the network and measures the response time.

Syntax:

```
ping destination
```

Example:

```
ping google.com
```

Sample Output:

```
64 bytes from google.com: icmp_seq=1 ttl=117 time=25 ms
```

Explanation:

This indicates that the system successfully communicated with Google's server and received a response in 25 milliseconds.

Send a specific number of packets:

```
ping -c 4 google.com
```

This sends only 4 ping requests instead of running continuously.

---

3. `netstat` (Network Statistics)

What does it do?

The `netstat` command displays network connections, routing tables, interface statistics, and listening ports.

Syntax:

```
netstat [options]
```

Example:

```
netstat -tuln
```

Explanation:

This command shows all active listening ports.

- `-t` → TCP connections
- `-u` → UDP connections
- `-l` → Listening ports
- `-n` → Display numerical addresses and port numbers

Sample Output:

```
tcp   0   0 0.0.0.0:22   0.0.0.0:*   LISTEN
```

This indicates that the SSH service is listening on port **22**.

---

4. `traceroute`

What does it do?

The `traceroute` command traces the route taken by packets from your computer to a destination host, showing each router (hop) along the path.

Syntax:

```
traceroute destination
```

Example:

```
traceroute google.com
```

Sample Output:

```
1  192.168.1.1     2 ms
2  10.10.0.1       8 ms
3  142.250.183.78  20 ms
```

Explanation:

The output lists the intermediate devices (hops) through which the data travels before reaching the destination.

Practical Example

Suppose you want to check your network status and internet connectivity:

View your network configuration:

```
ifconfig
```

Test connectivity to Google:

```
ping -c 4 google.com
```

Check active listening ports:

```
netstat -tuln
```

Trace the route to Google:

```
traceroute google.com
```

Conclusion

- **`ifconfig`** is used to view and configure network interface settings.
- **`ping`** is used to test network connectivity and measure response time.
- **`netstat`** is used to display network connections and listening ports.
- **`traceroute`** is used to identify the path packets take to reach a destination.

These commands are essential networking tools in Linux and are widely used for network administration, troubleshooting, and cybersecurity tasks.

Step 4 : Networking Basics

❖OSI Model Layers & Functions

OSI Model Layers & Functions

The **OSI (Open Systems Interconnection) Model** is a conceptual framework used to understand how data is transmitted between devices over a network. It divides the communication process into **seven layers**, where each layer performs specific functions and communicates with the layers above and below it.

---

1. Physical Layer (Layer 1)

Function:

The Physical Layer is responsible for transmitting raw bits of data over a physical medium such as cables, fiber optics, or wireless signals. It defines hardware specifications, connectors, voltage levels, and transmission rates.

Examples:

- Ethernet cables
- Hubs
- Repeaters
- Network connectors

---

2. Data Link Layer (Layer 2)

Function:

The Data Link Layer ensures reliable data transfer between directly connected devices. It organizes bits into frames, detects errors, and controls access to the transmission medium.

Examples:

- MAC Addresses
- Switches

3. Network Layer (Layer 3)

Function:

The Network Layer is responsible for logical addressing and routing data packets between different networks. It determines the best path for data delivery.

Examples:

- IP Addresses
- Routers

4. Transport Layer (Layer 4)

Function:

The Transport Layer provides end-to-end communication, ensuring that data is delivered accurately and in the correct order. It handles segmentation, flow control, and error recovery.

Examples:

- TCP (Transmission Control Protocol)
- UDP (User Datagram Protocol)

5. Session Layer (Layer 5)

Function:

The Session Layer establishes, manages, and terminates communication sessions between applications. It controls dialogue synchronization and session checkpoints.

Examples:

- Remote Procedure Calls (RPC)
- NetBIOS Sessions

6. Presentation Layer (Layer 6)

Function:

The Presentation Layer translates, encrypts, and compresses data so that it can be understood by the receiving application. It acts as a translator between the application and the network.

Examples:

- SSL/TLS Encryption
- JPEG
- GIF
- ASCII Encoding

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7. Application Layer (Layer 7)

Function:

The Application Layer provides network services directly to end users and application software. It enables users to access network resources and services.

Examples:

- HTTP/HTTPS
- FTP
- SMTP
- DNS
- Telnet

❖TCP/IP Protocol Suite

TCP/IP Protocol Suite

The **TCP/IP (Transmission Control Protocol/Internet Protocol) Protocol Suite** is a set of communication protocols used to connect devices over the Internet and other networks. It provides the rules and standards that enable computers to communicate and exchange data reliably. The TCP/IP model is the foundation of modern networking and is widely used in operating systems such as Windows, Linux, and macOS.

Layers of the TCP/IP Protocol Suite

The TCP/IP model consists of **four layers**, each responsible for specific networking functions.

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1. Application Layer

Function:

The Application Layer provides network services directly to user applications. It enables communication between software applications and the network.

Responsibilities:

- User interaction with network services
- File transfer
- Email communication
- Web browsing
- Name resolution

Examples of Protocols:

- **HTTP/HTTPS** – Used for web browsing
- **FTP** – Used for file transfer

---

2. Transport Layer

Function:

The Transport Layer provides end-to-end communication between devices. It ensures that data is delivered correctly and efficiently.

Responsibilities:

- Data segmentation and reassembly
- Error detection and recovery
- Flow control
- Reliable or unreliable delivery
- Port addressing

Protocols:

TCP (Transmission Control Protocol)

- Connection-oriented protocol
- Provides reliable data delivery
- Performs error checking and retransmission
- Ensures packets arrive in the correct order

UDP (User Datagram Protocol)

- Connectionless protocol
- Faster but less reliable than TCP
- Does not guarantee delivery or packet order
- Commonly used for streaming and online gaming

---

3. Internet Layer

Function:

The Internet Layer is responsible for logical addressing and routing packets across networks.

Responsibilities:

- Packet forwarding
- Routing
- Logical addressing
- Fragmentation and reassembly

Examples of Protocols:

- **IP (Internet Protocol)** – Provides logical addressing
- **ICMP** – Used for error reporting and diagnostics

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4. Network Access Layer

Function:

The Network Access Layer handles the physical transmission of data over the network medium and defines how data is sent between devices on the same network.

Responsibilities:

- Framing
- Physical addressing
- Error detection
- Access to transmission media

Examples:

- Ethernet
- Wi-Fi

❖DNS & HTTP/HTTPS Deep Dive

DNS & HTTP/HTTPS Deep Dive

Introduction

DNS (Domain Name System) and **HTTP/HTTPS (Hypertext Transfer Protocol / Hypertext Transfer Protocol Secure)** are two of the most important technologies that make the Internet work. DNS helps users locate websites by converting domain names into IP addresses, while HTTP and HTTPS enable communication between web browsers and web servers to exchange web content.

---

1. DNS (Domain Name System)

What is DNS?

The **Domain Name System (DNS)** is often called the **"phonebook of the Internet."** Humans prefer using easy-to-remember domain names such as **www.google.com**, whereas computers communicate using numerical IP addresses such as **142.250.183.78**. DNS translates domain names into IP addresses so that users can access websites without memorizing numeric addresses.

Functions of DNS

- Converts domain names into IP addresses.
- Helps users access websites using readable names.
- Reduces the need to remember complex IP addresses.
- Distributes requests across multiple servers.
- Supports email routing and other Internet services.

How DNS Works

When a user enters a website address into a browser, the following process occurs:

Step 1: User Request

The user types a domain name, such as:

```
www.example.com
```

into a web browser.

Step 2: DNS Resolver

The browser sends a query to a **DNS Resolver**, usually operated by the Internet Service Provider (ISP).

Step 3: Root DNS Server

If the resolver does not already have the answer cached, it contacts a **Root DNS Server**, which directs it to the appropriate Top-Level Domain server.

Step 4: Top-Level Domain (TLD) Server

The TLD server (such as `.com`, `.org`, or `.net`) identifies the authoritative name server responsible for the domain.

Step 5: Authoritative DNS Server

The authoritative server provides the correct IP address associated with the domain name.

Step 6: Response to User

The resolver returns the IP address to the browser, allowing it to establish a connection with the website.

Example

```
google.com  →  142.250.183.78
```

DNS converts the domain name into an IP address that computers can understand.

---

2. HTTP (Hypertext Transfer Protocol)

What is HTTP?

HTTP is the protocol used by web browsers and web servers to exchange information on the World Wide Web. It follows a **client-server model**, where the browser acts as the client and the website server acts as the server.

HTTP transfers web pages, images, videos, and other web resources.

Functions of HTTP

- Requests and delivers web pages.
- Transfers multimedia content.
- Enables communication between browsers and servers.
- Supports web applications and APIs.

How HTTP Works

Step 1: Browser Sends Request

Example:

```
GET /index.html HTTP/1.1
Host: example.com
```

Step 2: Server Processes Request

The web server receives and processes the request.

Step 3: Server Sends Response

Example:

```
HTTP/1.1 200 OK
Content-Type: text/html
```

The requested webpage is returned to the browser.

Step 4: Browser Displays Content

The browser renders the webpage for the user.

Common HTTP Methods

Method	Purpose
GET	Retrieves data from the server
POST	Sends data to the server
PUT	Updates existing resources
DELETE	Removes resources
PATCH	Partially updates resources
HEAD	Retrieves headers only
OPTIONS	Displays supported methods

HTTP Status Codes

Status Code	Meaning
200	OK
201	Created

Status Code	Meaning
301	Moved Permanently
400	Bad Request
401	Unauthorized
403	Forbidden
404	Not Found
500	Internal Server Error
503	Service Unavailable

---

3. HTTPS (Hypertext Transfer Protocol Secure)

What is HTTPS?

HTTPS is the secure version of HTTP. It uses **SSL/TLS (Secure Sockets Layer/Transport Layer Security)** to encrypt communication between the browser and the web server.

Websites using HTTPS are identified by:

```
https://
```

and usually display a padlock icon in the browser.

Functions of HTTPS

- Encrypts transmitted data.
- Protects sensitive information.
- Verifies website identity.
- Prevents eavesdropping and data tampering.
- Builds trust between users and websites.

How HTTPS Works

Step 1: Client Hello

The browser initiates a secure connection and sends supported encryption methods.

Step 2: Server Hello

The server selects an encryption method and sends its SSL/TLS certificate.

Step 3: Certificate Verification

The browser verifies the server's digital certificate using trusted Certificate Authorities (CAs).

Step 4: Session Key Exchange

Both parties establish a shared session key.

Step 5: Encrypted Communication

All subsequent data exchanged between the browser and server is encrypted.

Relationship Between DNS and HTTP/HTTPS

When a user visits a website:

1. The browser requests the domain name.
2. DNS translates the domain name into an IP address.
3. The browser connects to the web server using that IP address.
4. HTTP or HTTPS is used to exchange data between the browser and the server.
5. The webpage is displayed to the user.

IP Addressing, Subnetting, and NAT

Introduction

IP Addressing, Subnetting, and Network Address Translation (NAT) are fundamental concepts in computer networking. They help identify devices on a network, divide networks into smaller segments for efficient management, and enable multiple devices to share a single public IP address. Understanding these concepts is essential for network administration, troubleshooting, and cybersecurity.

---

1. IP Addressing

What is an IP Address?

An **IP (Internet Protocol) Address** is a unique numerical identifier assigned to each device connected to a network. It allows devices to communicate and exchange data over the Internet or local networks.

Functions of an IP Address

- Identifies devices on a network.
- Enables communication between hosts.
- Determines the source and destination of data packets.
- Supports routing across networks.

Types of IP Addresses

IPv4

IPv4 uses **32-bit addresses** divided into four octets separated by periods.

Example:

```
192.168.1.10
```

Characteristics:

- Supports approximately 4.3 billion addresses.
- Written in decimal notation.
- Most commonly used version of IP.

IPv6

IPv6 uses **128-bit addresses** written in hexadecimal format.

Example:

```
2001:0db8:85a3:0000:0000:8a2e:0370:7334
```

Characteristics:

- Provides a significantly larger address space.
- Improves routing efficiency.
- Designed to replace IPv4.

Public and Private IP Addresses

Public IP Address

- Assigned by an Internet Service Provider (ISP).
- Accessible over the Internet.
- Must be globally unique.

Private IP Address

Used within local networks and not directly accessible from the Internet.

Private IPv4 Ranges:

Class	Address Range
Class A	10.0.0.0 – 10.255.255.255
Class B	172.16.0.0 – 172.31.255.255

Class	Address Range
Class C	192.168.0.0 – 192.168.255.255

---

2. Subnetting

What is Subnetting?

Subnetting is the process of dividing a large network into smaller logical networks called **subnets**. This improves network performance, security, and efficient utilization of IP addresses.

Benefits of Subnetting

- Reduces network congestion.
- Improves security by isolating network segments.
- Conserves IP addresses.
- Simplifies network management.
- Enhances overall performance.

Subnet Mask

A **Subnet Mask** determines which portion of an IP address represents the network and which portion represents the host.

Example:

IP Address:

```
192.168.1.50
```

Subnet Mask:

```
255.255.255.0
```

CIDR Notation:

```
/24
```

This means:

- First 24 bits represent the network portion.
- Remaining 8 bits represent host addresses.

Example of Subnetting

Consider the network:

```
192.168.1.0/24
```

A `/24` network provides:

- Total Addresses: 256
- Usable Host Addresses: 254
- Network Address: 192.168.1.0
- Broadcast Address: 192.168.1.255

Dividing into Two `/25` Subnets:

Subnet	Address Range	Broadcast Address	Usable Hosts
192.168.1.0/25	192.168.1.1 – 192.168.1.126	192.168.1.127	126
192.168.1.128/25	192.168.1.129 – 192.168.1.254	192.168.1.255	126

---

3. NAT (Network Address Translation)

What is NAT?

Network Address Translation (NAT) is a technique used by routers to translate private IP addresses into public IP addresses and vice versa. It allows multiple devices within a private network to share a single public IP address.

Functions of NAT

- Conserves public IPv4 addresses.

- Provides a layer of privacy by hiding internal IP addresses.
- Enables Internet access for private networks.
- Simplifies network administration.

How NAT Works

Suppose three devices inside a home network have private IP addresses:

```
192.168.1.2
192.168.1.3
192.168.1.4
```

The router has a public IP address:

```
203.0.113.10
```

When these devices access the Internet:

1. The router replaces each private IP address with its public IP address.
2. It records the connection details in a translation table.
3. Responses from the Internet return to the router.
4. The router forwards the responses to the correct internal devices.

Types of NAT

Static NAT

- Maps one private IP address to one public IP address.
- Used when a device requires a fixed public address.

Dynamic NAT

- Maps private IP addresses to a pool of public IP addresses.
- Assigns public addresses as needed.

PAT (Port Address Translation)

- Also known as **NAT Overload**.
- Multiple devices share a single public IP address using different port numbers.
- Most commonly used in home and office networks.

Step 5 : Cryptography Basics

❖Symmetric vs Asymmetric Encryption

Symmetric vs Asymmetric Encryption in Cryptography

Introduction

Encryption is the process of converting readable data (plaintext) into an unreadable format (ciphertext) to protect it from unauthorized access. Cryptography mainly uses two types of encryption: **Symmetric Encryption** and **Asymmetric Encryption**. Both play an essential role in securing digital communication, protecting sensitive information, and ensuring data confidentiality.

---

1. Symmetric Encryption

What is Symmetric Encryption?

Symmetric Encryption is a type of encryption in which the **same key is used for both encryption and decryption**. The sender and receiver must possess the identical secret key to securely exchange information.

How It Works

1. The sender encrypts the plaintext using a secret key.
2. The encrypted data (ciphertext) is transmitted.
3. The receiver uses the same secret key to decrypt the ciphertext back into plaintext.

Example:

```
Plaintext → Secret Key → Ciphertext → Secret Key → Plaintext
```

Advantages of Symmetric Encryption

- Faster than asymmetric encryption.
- Efficient for encrypting large amounts of data.
- Requires less computational power.
- Suitable for real-time applications.

Disadvantages of Symmetric Encryption

- Secure key distribution is difficult.
- Both parties must share the same secret key.
- If the key is compromised, all encrypted data becomes vulnerable.
- Not practical for communication with a large number of users.

Common Symmetric Encryption Algorithms

- **AES (Advanced Encryption Standard)**
- **DES (Data Encryption Standard)**
- **3DES (Triple DES)**
- **Blowfish**
- **Twofish**
- **ChaCha20**

Applications of Symmetric Encryption

- Disk and file encryption.
- Database encryption.
- VPN connections.
- Wireless network security.
- Backup encryption.

---

2. Asymmetric Encryption

What is Asymmetric Encryption?

Asymmetric Encryption uses **two mathematically related keys**:

- A **Public Key** – Shared openly with others.
- A **Private Key** – Kept secret by the owner.

Data encrypted with one key can only be decrypted using the corresponding paired key.

How It Works

1. The sender encrypts the plaintext using the recipient's public key.
2. The ciphertext is transmitted over the network.
3. The recipient decrypts the ciphertext using their private key.

Example:

```
Plaintext → Public Key → Ciphertext → Private Key → Plaintext
```

Advantages of Asymmetric Encryption

- Eliminates the need to share secret keys.
- Provides secure key exchange.
- Supports digital signatures and authentication.
- Enhances non-repudiation.

Disadvantages of Asymmetric Encryption

- Slower than symmetric encryption.
- Requires more computational resources.
- Less efficient for encrypting large volumes of data.

Common Asymmetric Encryption Algorithms

- **RSA**
- **ECC (Elliptic Curve Cryptography)**
- **Diffie–Hellman** (key exchange)
- **DSA (Digital Signature Algorithm)**
- **ElGamal**

Applications of Asymmetric Encryption

- Secure web communication (HTTPS/TLS).
- Digital signatures.
- Email encryption.
- Secure key exchange.
- Public Key Infrastructure (PKI).

❖Hashing (MD5, SHA256).

Hashing (MD5, SHA-256)

Introduction

Hashing is a cryptographic process that converts data of any size into a fixed-length value called a **hash value** or **message digest**. Unlike encryption, hashing is a **one-way process**, meaning the original data cannot be recovered from the hash value. Hashing is widely used for data integrity verification, password storage, and digital signatures.

What is Hashing?

Hashing takes an input, such as a file, password, or message, and generates a unique string of characters representing that data.

Example:

```
Input: Hello World
↓
Hash Function
↓
Output: Fixed-Length Hash Value
```

Even a small change in the input produces a completely different hash value.

Characteristics of Hash Functions

- **Deterministic:** The same input always produces the same output.
- **Fixed-Length Output:** Generates a hash of a specific length regardless of input size.
- **One-Way Function:** Original data cannot be reconstructed from the hash.
- **Fast Computation:** Efficiently generates hash values.
- **Collision Resistant:** Different inputs should not produce the same hash output.
- **Sensitive to Input Changes:** Even a one-character change drastically changes the hash.

---

1. MD5 (Message Digest Algorithm 5)

What is MD5?

MD5 is a widely known cryptographic hash function developed by **Ronald Rivest** in 1991. It produces a **128-bit (16-byte)** hash value, usually represented as a 32-character hexadecimal number.

How MD5 Works

1. The input message is divided into blocks.
2. Mathematical operations are performed on each block.
3. A 128-bit hash value is generated.

Example:

Input:

```
Hello World
```

MD5 Hash:

```
b10a8db164e0754105b7a99be72e3fe5
```

Advantages of MD5

- Fast and efficient.
- Easy to implement.
- Useful for file integrity checks.

Disadvantages of MD5

- Vulnerable to collision attacks.
- No longer considered secure for cryptographic purposes.
- Not recommended for password storage or digital signatures.

Applications of MD5

- File integrity verification.
- Checksum generation.
- Detecting accidental file corruption.

---

2. SHA-256 (Secure Hash Algorithm 256)

What is SHA-256?

SHA-256 is a member of the **SHA-2 family** developed by the **National Security Agency (NSA)** and published by **NIST**. It generates a **256-bit (32-byte)** hash value represented as a 64-character hexadecimal string.

How SHA-256 Works

1. The input data is processed in blocks.
2. Complex mathematical transformations are applied.
3. A 256-bit hash value is generated.

Example:

Input:

```
Hello World
```

SHA-256 Hash:

```
a591a6d40bf420404a011733cfb7b190
d62c65bf0bcda32b57b277d9ad9f146e
```

Advantages of SHA-256

- Highly secure and collision-resistant.
- Suitable for modern cryptographic applications.
- Widely trusted and adopted.
- Resistant to known practical attacks.

Disadvantages of SHA-256

- Slower than MD5.
- Requires more computational resources.

Applications of SHA-256

- Password hashing (with salting and key stretching).
- Digital signatures.
- SSL/TLS certificates.
- Blockchain technologies such as Bitcoin.
- File integrity verification.

MD5 vs SHA-256

Feature	MD5	SHA-256
Full Form	Message Digest Algorithm 5	Secure Hash Algorithm 256
Output Size	128 bits	256 bits
Hexadecimal Length	32 characters	64 characters
Security Level	Weak	Strong

Feature	MD5	SHA-256
Collision Resistance	Poor	Excellent
Speed	Faster	Slightly slower
Recommended for Cryptography	No	Yes
Common Uses	Checksums, file verification	Password security, digital signatures, blockchain

Practical Example

Suppose two users download the same file from the Internet. The website provides a SHA-256 hash value:

```
Expected SHA-256:
2cf24dba5fb0a30e26e83b2ac5b9e29e
1b161e5c1fa7425e73043362938b9824
```

After downloading the file, users generate the hash on their systems.

- If the generated hash matches the expected hash, the file is authentic and unchanged.
- If the hashes differ, the file may be corrupted or modified.

❖Hands-on: Encrypt and Decrypt messages using OpenSSL

➢Encrypt messages using OpenSSL

Created a text file named `secret.txt`.

➢ Decrypt messages using OpenSSL

OpenSSL Encryption & Decryption

Objective:
To understand basic cryptography by encrypting and decrypting a message using OpenSSL.

Procedure:

1. Created a text file named `secret.txt`.
2. Encrypted the file using AES-256-CBC encryption.
3. Protected the encrypted file using a password.
4. Decrypted the file using the same password.
5. Verified that the original message was successfully recovered.

Result:
The message was encrypted and decrypted successfully using OpenSSL.

Step 6 : Tool Familiarization

➢ Wireshark (packet capture).

Wireshark

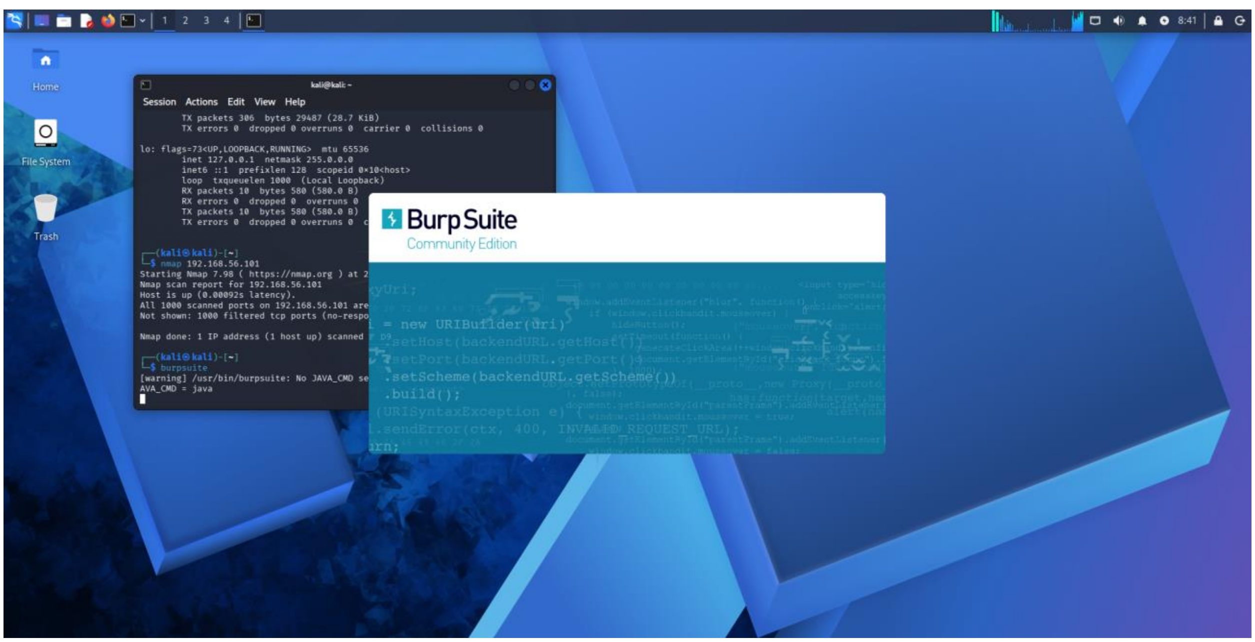
Used for capturing and analyzing network packets.

➢ Nmap (network scanning).

Nmap

Used for discovering hosts and scanning open ports.

➢ Burp Suite (web proxy).



Burp Suite

Used for web application security testing.

- ➢ Netcat (network debugging).

Netcat

Used for network debugging and communication testing.

END